Arduino
Introdução

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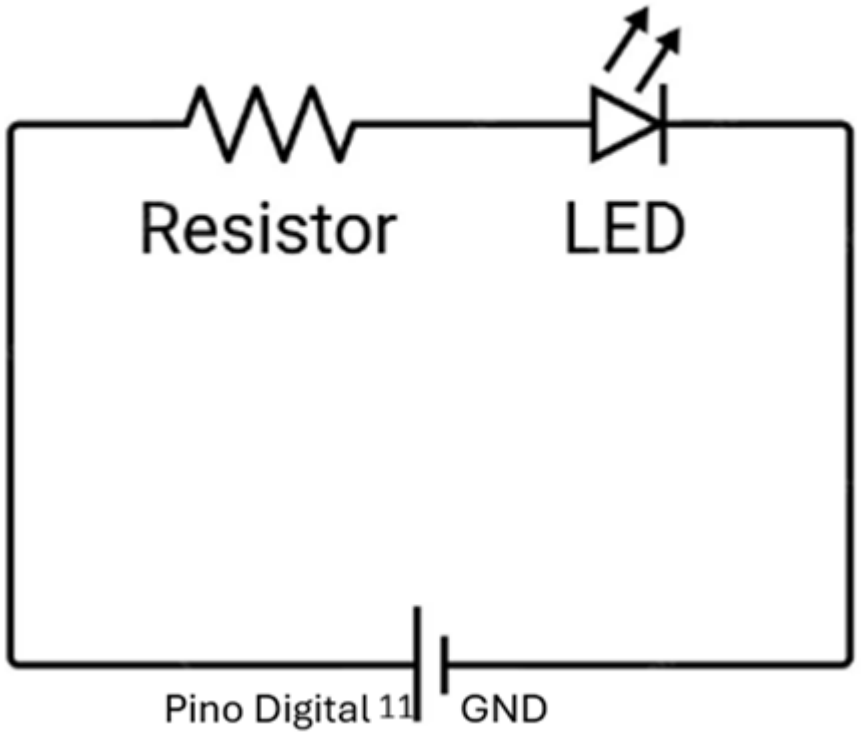
Parte 2

Saída Analógica

Pino Digital com ~ (PWM)

Usar Led Branco



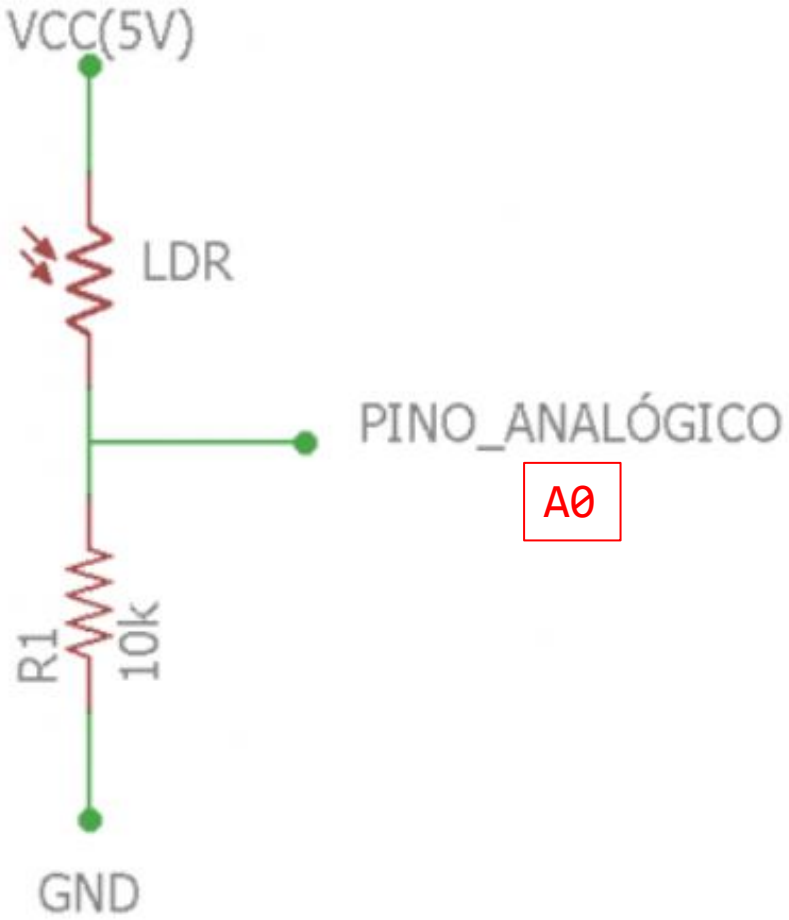

```
const int pinoLed = 11;

void setup() {
  pinMode(pinoLed,OUTPUT);
}

void loop() {
  // Saída PWN
  analogWrite(pinoLed,125);
}
```

Define valor da saída analógica PWM

Led acenderá próximo a metade da intensidade.
Interfavo entre 0 a 255

Entrada e Saída Analógica

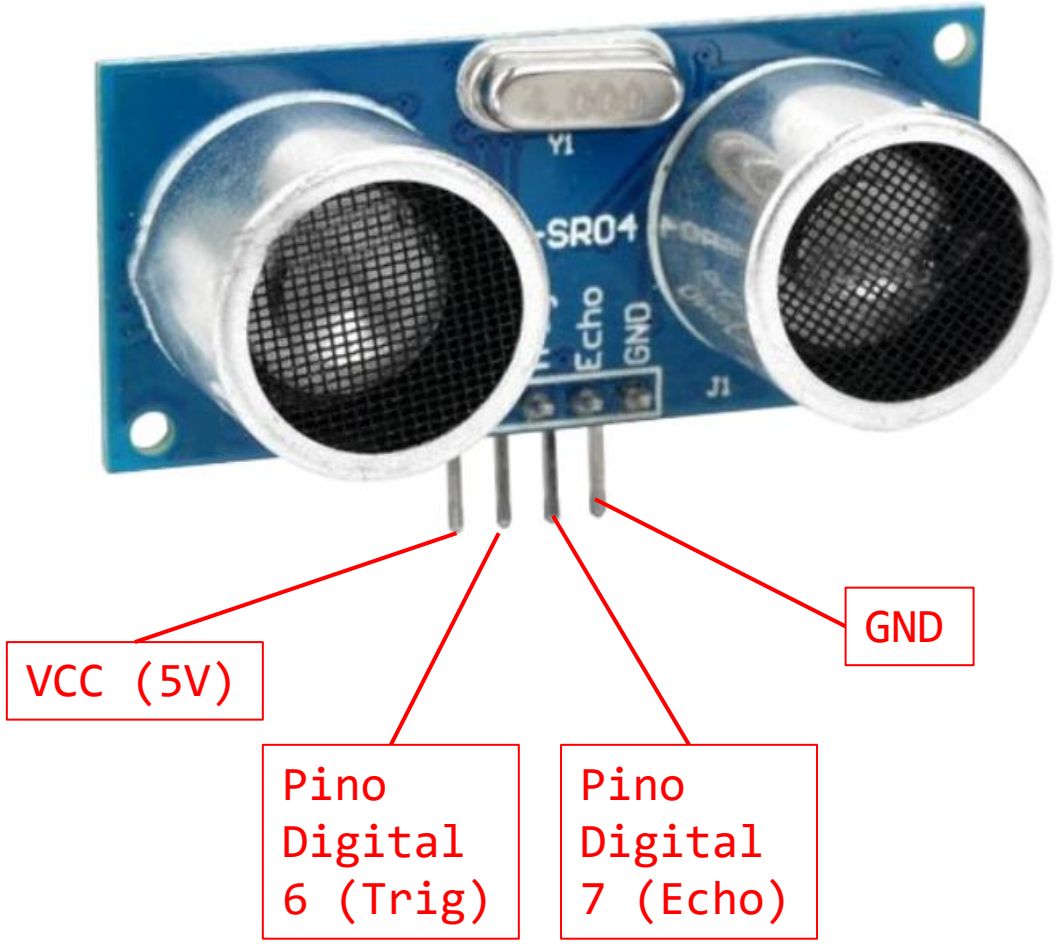
Junto com circuito anterior, remontar o circuito com sensor de luminosidade.




```
const int pinoLed = 11;  // saída PWM
const int pinoSensorLuz = A0; // entrada analógica
void setup() {
  pinMode(pinoLed,OUTPUT);
  pinMode(pinoSensorLuz,INPUT);
}

void loop() {
  int leitura = analogRead(pinoSensorLuz);
  int saidaLed = map(leitura,0,1023,0,255);
  analogWrite(pinoLed,saidaLed);
}
```

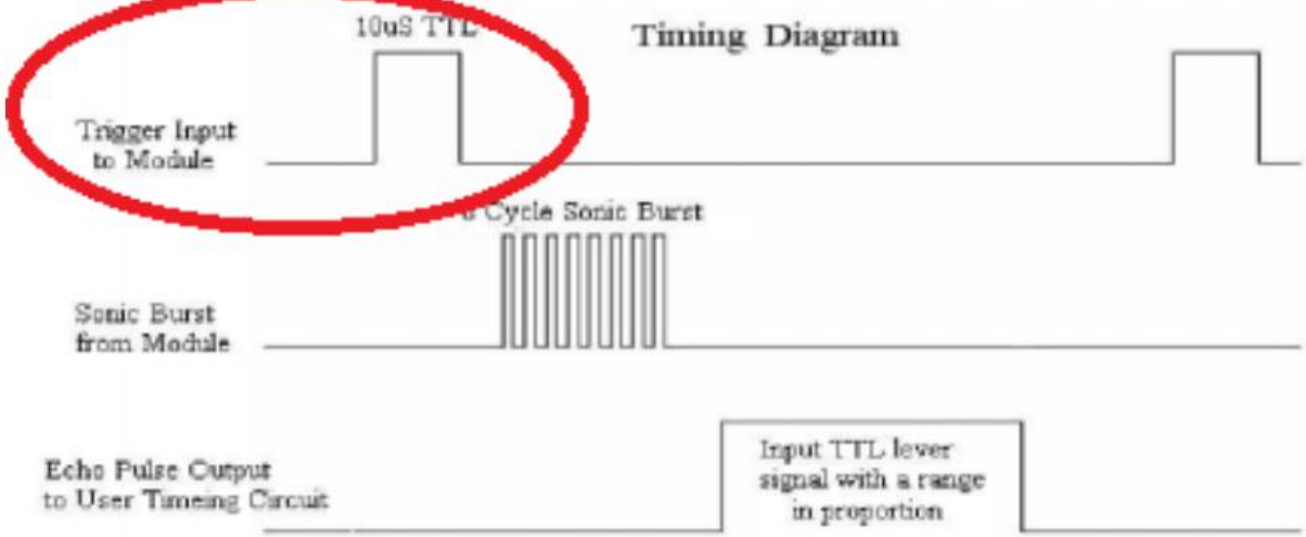
Transforma valor no intervalo [0,1023] para o intervalo [0,255]

Sensor de Distância



Timing diagram

The Timing diagram is shown below. You only need to supply a short 10uS pulse to the trigger input to start the ranging, and then the module will send out an 8 cycle burst of ultrasound at 40 kHz and raise its echo. The Echo is a distance object that is pulse width and the range in proportion .You can calculate the range through the time interval between sending trigger signal and receiving echo signal. Formula: uS / 58 = centimeters or uS / 148 =inch; or: the range = high level time * velocity (340M/S) / 2; we suggest to use over 60ms measurement cycle, in order to prevent trigger signal to the echo signal.



Sensor de Distância

```
const int pinoTrig = 6;
const int pinoEcho = 7;
void setup() {
  pinMode(pinoTrig,OUTPUT);
  pinMode(pinoEcho,INPUT);
  Serial.begin(9600);
}
void loop() {
  int distancia = medeDistancia(pinoTrig,pinoEcho);
  Serial.println(distancia);
  delay(1000);
}
int medeDistancia(int pinoTrig,int pinoEcho){
  // Gera pulso de 10 microsegundos no Trig
  digitalWrite(pinoTrig,LOW);
  delayMicroseconds(2);
  digitalWrite(pinoTrig,HIGH);
  delayMicroseconds(10);
  digitalWrite(pinoTrig,LOW);
  // Calcula distancia medindo a largura do pulso em Echo
  return pulseIn(pinoEcho,HIGH)/58;
}
```

Alerta Sonoro Conforme Distância

```
const int pinoTrig = 6;
const int pinoEcho = 7;
const int pinoBuzina = 8;
void setup() {
  pinMode(pinoTrig,OUTPUT);
  pinMode(pinoEcho,INPUT);
  pinMode(pinoBuzina,OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int distancia = medeDistancia(pinoTrig,pinoEcho);
  Serial.println(distancia);
  if(distancia <= 10) {
    tone(pinoBuzina,400);
  } else if(distancia <= 20) {
    tone(pinoBuzina,300);
  } else {
    noTone(pinoBuzina);
  }
  delay(1000);
  noTone(pinoBuzina);
  delay(1000);
}
```

Junto com circuito anterior, montar sinalizador sonoro (Buzzer). ATENÇÃO PARA A POLARIDADE.

GND

Pino Digital 8

Servo-Motor

Conexões:
VCC (5V) – fio vermelho
GND – fio marrom
Pino Digital 9 – fio laranja

Angulação Útil 10 a 170 graus

VCC (5V)

Pino Analógico A0

GND

Servo-Motor

```
#include <Servo.h> // Biblioteca para servo motor, já vem instalada

Servo motor;  // Objeto para controlar o servo motor
const int potenciometro = A0;

void setup() {
  motor.attach(9);  // Informa qual pino será usado para o servo declarado anteriormente.
  motor.write(10); //Informa ao servo que deverá se posicionar a 10 graus
  pinMode(potenciometro,INPUT);
}

void loop() {
  int leitura = analogRead(potenciometro); // Obtém a leitura do potenciometro
  leitura = map(leitura,0,1023,10,170); // Transforma de [0,1013] para [10,170] (ângulo motor)
  motor.write(leitura); // Informa ao servo a posição em função da leitura do potenciometro
  delay(100);
}
```